

LLM Evaluation Metrics

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What is Harness Engineering?

Harness Engineering is the practice of building a **testing and evaluation framework** (harness) to systematically measure the **quality, reliability, and performance** of AI models

1. **Correctness** ex: Ground truth- Tokyo, IIm: Kyoto (Exact match, accuracy, human review, LLM-as-a-judge (with caution))
2. **Relevance** - Fully relevant, partially relevant, not relevant
3. **Groundedness** - supported by retrieved context, Percentage of grounded responses
4. **Hallucination Rate** - model invents information.
5. **Faithfulness** - Everything stated is supported by retrieved docs.
6. **Context precision** - Not ideal
7. **Context Recall** -
8. **Safety** - Toxicity, Bias, PII (Personally identifiable information) leakage, Prompt injection resistance, Jailbreak resistance, Harmful content generation.
9. **Latency** - API Response time, Model inference time, vector search time, DB time
10. **Cost** - Input token, Output token, Embedding cost, API cost, Gpu cost
11. **User experience** - Ratings, Follow-up question, Task completion rate, Escalation to humans, User surveys.
12. **Robustness** - Typos, Ambiguous prompts, incomplete inputs, Adversarial prompts, Noisy data.
13. **Reasoning Quality** - Chain-of-Thought correctness

Common Automated Metrics

These are useful for specific tasks but have limitations for open-ended generation.

Metric	Best For	Limitation
Exact Match	QA	Too strict—different wording may still be correct
BLEU	Translation	Focuses on word overlap rather than meaning
ROUGE	Summarization	Rewards overlap with a reference summary
BERTScore	Semantic similarity	Better captures meaning but still needs reference answers

BLEU - Bilingual Evaluation Understudy

ROUGE - Recall Oriented Understudy for Gisting Evaluation

MRR - Mean Reciprocal Rank

NDCG - Normalized Discounted Cumulative Gain

Comparison Table

Feature	MRR	NDCG
Focus	Rank of the first relevant item only	Quality of the entire top-\$k\$ list
Relevance Type	Binary (Relevant / Not Relevant)	Graded (e.g., 0 to 4 scale)
Sensitivity	Insensitive to ranks after the first hit	Sensitive to the order of all relevant items
Ideal Use Case	Factoid Q&A, Known-item search, Tool routing	Recommendation systems, Search engines, RAG with chunk grading

Typical Harness Architecture

Test Dataset



Prompt / RAG / Agent



LLM Response



Evaluation Metrics



Report / Dashboard

10. Benchmark Datasets

Popular benchmarks include:

Benchmark	Purpose
MMLU	General knowledge and reasoning
HumanEval	Code generation
GSM8K	Grade-school math reasoning
BIG-bench	Diverse reasoning tasks
ARC	Scientific reasoning
HellaSwag	Commonsense inference
TruthfulQA	Hallucination and truthfulness
MT-Bench	Multi-turn conversational quality
SWE-bench	Software engineering task performance
AgentBench	Autonomous agent evaluation
GAIA	Real-world multi-step agent tasks

11. Open-Source Evaluation Frameworks

Several mature frameworks simplify building evaluation pipelines:

Framework	Best For
DeepEval	LLM evaluation with custom metrics and LLM-as-a-judge
LangSmith	Tracing, debugging, and evaluation of LangChain applications
OpenAI Evals	Regression testing and benchmark creation
Ragas	Retrieval-Augmented Generation (RAG) evaluation
TruLens	Feedback functions, RAG evaluation, and observability
MLflow Evaluate	Experiment tracking and model evaluation
Promptfoo	Prompt testing, regression testing, and security checks
HELM	Standardized benchmarking across models